

Chethan Chakradhar Mangala

Senior Data Analyst / AI/ML Engineer

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PROFESSIONAL SUMMARY:

- Accomplished professional with 10 years of experience in designing sophisticated **algorithms** for **incident alert identification and extraction**, reducing redundancy through **pattern recognition rules**.
- Extensive expertise in **exploratory data analysis (EDA)** using **Python** and **R**, calculating **descriptive statistics**, detecting **outliers**, and performing **factor analysis** to derive meaningful insights.
- Proven track record of leveraging **Spark**, **Python**, **R**, and various **machine learning methods** for **classifications, regressions, and dimensionality reduction**, aligning strategies with domain knowledge and business objectives.
- Innovative thinker with a focus on scalable solutions using **machine learning**, **data mining**, and **statistical techniques** to address complex business problems.
- Expertise in **Data Extraction, Modeling, Wrangling, Mining, Statistical Analysis, Machine Learning, and Visualization** across **Retail, Banking, and Manufacturing** domains.
- Experience in **Artificial Intelligence (AI)** techniques such as **Natural Language Processing (NLP)**, **Large Language Models (LLM)**, **Computer Vision**, and **machine learning algorithms** used to process and clean raw data.
- Integrated **Docker containers** with **Kubernetes** to orchestrate and scale a **high-availability web application**.
- Utilized **GCP resources** namely **BigQuery, Cloud Composer, Compute Engine, Kubernetes Cluster, and GCP Storage Buckets** for building the production **ML infrastructure**.
- Expertise in building **data pipelines** for automating the data process and ensuring consistency in processing.
- Solid understanding of **Data Modelling, Data Collection, Data Cleansing, Data Warehouse/Data Mart Design, Snowflake, ETL, BI, OLAP, and Client/Server applications**.
- Proficient in using data handling libraries such as **Pandas, NumPy, NLTK, TextBlob, Gensim, and Scikit-learn**.
- Proficient in visualizing the data with the help of **Matplotlib, Seaborn, and Plotly**.
- Expertise in **statistical analysis** of the data with hypothesis testing methods such as **One-sample Z-test, T-test, Two-sample hypothesis testing, Chi-square testing, and ANOVA**.
- Design, develop and implemented **GenAI, LLM, and NLP models** using **Machine Learning (ML Infrastructure)**.
- Sound expertise in **web scraping** using **Python** packages such as **requests, BeautifulSoup (bs4), lxml, and Selenium**.
- Utilized **GenAI models** to generate new creative text formats and answer all questions.
- Specialized in **Tableau Servers, ETL, Teradata, and other EDW data integrations and developments** as a **Programmer Analyst**.
- Understanding of various **Machine Learning (ML Infrastructure) algorithms** such as **Linear Regression, Lasso & Ridge, Logistic Regression, KNN, Naive Bayes, Decision Tree, Random Forest,**

K-means, DBSCAN, XGBoost, AdaBoost, Gradient Descent, Isolation Forest, Gaussian Mixture Models, SVM, SVC, and Matrix Factorization.

- Ability to handle NLP with the help of **NLTK, Gensim, Large Language Models (LLM), Hugging Face, TextBlob**, etc.
- **Python programming** using **OOPS** style by facilitating **Inheritance, Polymorphism, Encapsulation, and Abstraction**.
- Skilled in **Data Parsing, Data Manipulation, and Data Preparation** with methods including describing data contents.
- Skilled in **fine-tuning pre-trained LLMs** to adapt them to specific tasks and domains, leveraging **transfer learning** and **domain adaptation**.
- Strong foundation in **NLP with generative models** and **text generation**.
- Experience in **Agile/SCRUM methodologies, Jira**, and coordinating **user stories** and **timelines**.
- Strong communicator, adept at translating complex **data insights** for **technical and non-technical stakeholders**.

EDUCATION:

Bachelors in Jawaharlal Nehru Technological University

June 2013

TOOLS AND TECHNOLOGIES:

Programming Languages	Python, JAVA, JavaScript, jQuery, ReactJS, Next.js, HTML, CSS, C, C++, Angular, R, Impala, Hive, SQL
Statistical Methods	Statistical Inference, Hypothesis Testing, Confidence Intervals, p-values, Statistical Significance, Probability Distributions, Descriptive Statistics, Correlation and Covariance, Sampling Techniques, ANOVA, Chi-Square Tests, Bayes' Theorem, Cross-Validation, Time Series Analysis, Auto-correlation, Statistical Modeling, Experimental Design, Central Limit Theorem, Law of Large Numbers, Residual Analysis, Multivariate Analysis
Machine Learning	Supervised Learning, Unsupervised Learning, Model Evaluation Metrics, Cross-Validation, Feature Engineering, Hyperparameter Tuning, Overfitting & Regularization (L1, L2), Ensemble Methods, Decision Trees, Random Forest, Gradient Boosting (XGBoost, LightGBM), Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Naive Bayes, Regression (Linear, Logistic), Classification Algorithms, Clustering (K-Means, Hierarchical), Dimensionality Reduction (PCA, t-SNE), Time Series Forecasting, ARIMA, SARIMA, Holt-Winters Exponential Smoothing, Prophet, LSTM for Time Series, Model Interpretability (SHAP, LIME), Recommendation Systems, Scikit-learn
Deep Learning	Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Autoencoders, Transfer Learning(VGG, ResNet, InceptionNet, MobileNet), Attention Mechanism, Object Detection (YOLO, Faster R-CNN), OCR, Image Segmentation (U-Net, Mask R-CNN), Activation Functions (ReLU, Sigmoid, Tanh, Softmax), Optimization Algorithms (SGD, Adam, RMSprop), Loss Functions (Cross-Entropy, MSE), Regularization (Dropout, Batch Normalization), Model Deployment, TensorFlow, Keras, PyTorch
Natural Language Processing (NLP)	Text Preprocessing (Tokenization, Lemmatization, Stemming), Bag of Words (BoW), TF-IDF, Word Embeddings (Word2Vec, GloVe, FastText), Sequence-to-Sequence Models, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Bidirectional LSTM (Bi-LSTM), Attention Mechanism, Encoder-Decoder Models, Gated Recurrent Units (GRU), Transformer Models (BERT, GPT, T5), Named Entity Recognition (NER), Part-of-Speech Tagging, Sentiment Analysis, Text Classification, Machine Translation, Text Summarization, Question Answering Systems, Language

	Modeling, Word and Sentence Similarity, POS Tagging, Dependency Parsing, Text Generation, Sequence Labeling, Topic Modeling (LDA), SpaCy, NLTK, Hugging Face
Generative AI	LLM's, Ollama, Langchain, Langsmith, Agentic AI, Fine-tuning Techniques (LoRA, QLoRA), Local LLMs (Mistral, Llama, Gemma, etc.), Inference APIs (Groq, Nvidia NIM, etc.), Retrieval-Augmented Generation (RAG), Graph Databases(Neo4j), Vector Databases (Faiss, Chroma, Pinecone), Embedding Models(Mistral, GoogleAI Embedding Models), Prompt Engineering, Prompt Templates, Prompt Versioning, Document Chunking & Indexing, Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Conditional GANs, CycleGAN, StyleGAN, Deep Convolutional GANs (DCGAN), Transformer-based Generative Models (GPT, T5), Text-to-Image Generation (DALL-E, CLIP), Image-to-Image Translation, Neural Style Transfer, Chatbots, AI Search Engines, Semi-Supervised Learning, Data Augmentation, Zero-Shot Learning, Self-Supervised Learning, Reinforcement Learning for Generation, Diffusion Models, Latent Space Exploration
Experimentation	AB Testing Frameworks, Experimental Design, Control vs Test Group Analysis, p-value Interpretation
MLOps & LLMops	CI/CD for ML, Model Deployment (REST API, Batch), Model Monitoring, Model Versioning, Model Retraining Pipelines, ML Pipelines (MLflow, Airflow, Kubeflow), Experiment Tracking (MLflow, Weights & Biases), Data Versioning (DVC), Logging and Alerting (Prometheus, Grafana), Containerization (Docker), Orchestration (Kubernetes), Feature Stores (Feast), Cloud ML Platforms (AWS SageMaker, AWS BedRock, GCP AI Platform, Google Vertex AI, Azure AI) LLM Response Evaluation (TruLens, PromptLayer), Model Registry, Token Usage Tracking, Scalable LLM Deployment (TGI, vLLM)
Big Data:	Hadoop, Hive, HBase, Apache Spark, Scala, Databricks, Kinesis, Pig, Sqoop.
Amazon Web Services:	EC2, Lambda, Sage Maker, Bedrock, EMR, S3, Glue, Redshift, MKS, Kinesis, Quick Sight, API Gateway, Athena, Lex, Recognition, CI/CD, Code Commit, DynamoDB, transcribe, Cloud Formation, Cloud Watch, Glacier, IAM
Database Servers:	MySQL, Mongo Microsoft SQL server, SQLite, Red Shift, RDS, PostgreSQL, SQLAlchemy, Mongo DB, Teradata
Product Analytics	Funnel Conversion Analysis, User Retention Metrics, Cohort Analysis, Churn Prediction, A/B Testing, Hypothesis Testing, SQL-Based Behavioral Queries
Other Tools & Technologies:	Git, GitHub, GitLab, Docker, Docker Compose, Kubernetes, VS Code, Jupyter Notebook, Google Colab, CUDA Toolkit, Postman, Swagger, REST APIs, Linux/Unix Command Line, Bash Scripting, Conda, Virtualenv, Makefile, YAML, JSON, Terraform, Nginx, Redis, Jenkins, Power BI, Tableau, Excel, Google Sheets, Notion, Trello, Slack, Figma, QNXT

PROFESSIONAL EXPERIENCE:

Client: Verizon, USA

June 2024 – Present

Role: Sr. Data Analyst

Responsibilities:

- **Created parameters, action filters, and calculated sets in Tableau** to design **interactive dashboards, worksheets, and reports** for actionable business insights.
- **Extracted, cleaned, and analyzed large datasets** using **Google SQL, MySQL, and Snowflake**, ensuring data accuracy, integrity, and performance optimization.
- **Architected a real-time marketing KPI suite in Tableau** that integrated 5+ data sources (Snowflake, Teradata), leading to a **20% improvement in campaign ROI** through faster decision-making.
- **Developed ETL pipelines** for data migration and transformation into **Snowflake Data Warehouse**, improving reporting efficiency and accessibility.
- **Applied statistical techniques and ML algorithms** (Regression, Classification, Clustering, Anomaly Detection) to **identify business trends, customer patterns, and predictive insights**.
- **Designed and published Tableau Server dashboards**, including **report scheduling, automation, and stakeholder-driven reporting solutions**.
- **Used Python libraries (NumPy, Pandas, Matplotlib, Seaborn, SciPy)** for **data manipulation, visualization, and exploratory data analysis (EDA)**.
- **Implemented NLP techniques** (sentiment analysis, text classification, entity recognition) to generate insights from unstructured text data such as customer feedback.
- **Worked on Big Data platforms (Hadoop, Spark SQL, MapReduce)** to process, query, and analyze **large-scale datasets** for reporting and insights generation.
- **Applied Generative AI and LLM-based models (GPT, BERT)** to support **document summarization, chatbot building, and legal document analysis**, reducing manual review effort by **30%**.
- **Developed and maintained data visualizations in Tableau, Power BI, and Splunk**, delivering insights for **business, operations, and risk monitoring**.
- **Configured GCP services (BigQuery, DataProc, Storage) and Azure Data Factory pipelines**, ensuring **scalable data ingestion and analytics workflows**.
- **Collaborated with cross-functional teams** (architects, business analysts, product owners) to translate requirements into **data-driven solutions**.
- **Documented analysis workflows, dashboards, and KPIs**, ensuring reusability and alignment with **Agile/Scrum project management practices**.
- **Communicated complex insights clearly to stakeholders**, tailoring **AI/ML-driven results** into simple, actionable recommendations for **business decision-making**.

Environment: SQL (Google SQL, MySQL), Snowflake, GCP (BigQuery, DataProc, Storage), Azure Data Factory, Hadoop, Apache Spark (Spark SQL), MapReduce, Python, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Scikit-learn, NLP, LLMs (GPT, BERT), GenAI, Tableau (Desktop, Server), Power BI, Splunk, ETL Pipelines, Regression, Classification, Clustering, Anomaly Detection, Sentiment Analysis, Agile (Scrum).

Client: BCBS, USA

Oct 2022-May 2024

Role: Data Analyst / AI/ML Engineer

Responsibilities:

- **Analyzed and prepared data** by identifying patterns using historical models, collaborating with Senior Data Scientists to derive actionable insights.

- Performed **data manipulation, normalization, and predictive modeling** to improve efficiency and accuracy using **Python** and **R**.
- Designed and developed a **call quality control system** using **Python**, **Scikit-learn**, and **Machine Learning (Random Forest Regressor)**.
- Conducted **feature selection** with **GridSearchCV** to optimize model performance by identifying key predictors.
- Performed **data cleaning and preprocessing** using **Pandas** and **NumPy**, ensuring data quality and consistency.
- Applied **standardization techniques** to normalize call quality metrics across datasets.
- Built **Scikit-learn modeling pipelines** to automate workflows and increase reproducibility.
- Leveraged **SciPy** for **statistical analysis and hypothesis testing** to uncover correlations and causations.
- Implemented **continuous testing and maintenance** to ensure system reliability and robustness.
- Conducted **exploratory data analysis (EDA)** to detect outliers, anomalies, and hidden trends.
- Authored **documentation and training materials** to enable stakeholders' adoption of AI/ML solutions.
- Developed **real-time streaming applications** using **PySpark**, **Apache Flink**, **Kafka**, **Hive** on a **Hadoop Cluster**.
- Built **ETL pipelines** with **Python** and **Snowflake SQL** for scalable data ingestion and transformation.
- Designed **Flink pipelines** to consume and process streaming data from **Kafka**, applying business logic and transformations.
- Automated ingestion to **Hive** and **HDFS** from **Oracle** and **SQL Server** using **Sqoop**.
- Configured and automated **AWS services (EC2, S3, EBS, Lambda)** and deployed solutions on **Azure** for cloud-based ML applications.
- Applied **Azure Lambda** for real-time **data validation, filtering, and transformations** from **DynamoDB**.
- Worked with **Google Cloud Platform (GCP)** services such as **Compute Engine**, **Cloud Load Balancing**, **Cloud SQL**, **Stackdriver Monitoring**, **Cloud Deployment Manager**.
- Built **MapReduce** and **Spark Python modules** for **machine learning and predictive analytics** in Hadoop on **Azure**.
- Utilized **Spark**, **Hadoop**, **HBase**, **Kafka**, **Spark Streaming**, **Caffe**, **TensorFlow**, and **MLOps pipelines** for large-scale ML infrastructure, covering **classification, regression, and dimensionality reduction**.
- Performed **time series analysis** and built **interactive dashboards** with **Tableau** to visualize KPIs, trends, and anomalies.

Environment: SQL (Snowflake SQL, Google SQL, MySQL, Oracle, SQL Server), Snowflake, Azure Cloud, Azure Lambda, Azure Data Lake, Databricks, GCP (Compute Engine, Cloud Load Balancing, Cloud SQL, Stackdriver Monitoring, Cloud Deployment Manager), AWS (EC2, S3, EBS, Lambda), Hadoop Cluster, HDFS, Hive, HBase, Sqoop, Apache Spark, PySpark, Spark Streaming, Apache Flink, Kafka, Python, R, Pandas, NumPy, Scikit-learn, SciPy, TensorFlow, Caffe, MLOps, Random Forest Regressor, Time Series Analysis, GridSearchCV, Tableau.

Client: Visa, Philadelphia, PA

June 2021 – Sep 2022

Role: Sr. Data scientist

Responsibilities:

- Worked with large amounts of **structured and unstructured data**.
- Knowledge in **Machine Learning (ML Infrastructure)** concepts (**Generalized Linear Models, Regularization, Random Forest, Time Series Models**, etc.)

- Used **GenAI** to monitor systems that track the performance of deployed models, detecting anomalies or drifts in **data distribution** that may affect model accuracy.
- Responsible for building an **Azure Cloud Enterprise Data Platform**, including establishing connections between **Azure Resources (ADF, Databricks, ADLS GEN2, Storage layer access for ADF)**.
- Worked with **Business Intelligence (BI) tools** and visualization tools such as **BusinessObjects, Tableau, Chart IO**, etc.
- Used **ML (Reinforcement)** to monitor the performance of deployed models in real-time, tracking key metrics such as **accuracy, latency, and throughput**.
- Deployed **GUI pages** by using **JSP, JSTL, HTML, DHTML, XHTML, CSS, JavaScript, AJAX** and **Python**.
- Utilized **GCP's Machine Learning** libraries like **TensorFlow** for advanced feature engineering tasks.
- Utilized **GPT models** to develop **Natural Language Processing (NLP)** solutions, including **chatbots** and **text tools**.
- Utilized **PostgreSQL** and **MySQL** for **data preprocessing** tasks such as data cleaning, filtering, joining.
- Imported data from **Azure S3** into **Spark RDD**, performed **transformations and actions** on RDDs. Implemented **ElasticSearch** on **Hive Data Warehouse** platform and worked with **Elastic MapReduce**, and setup **Hadoop environment** in **Azure EC2 Instances**.
- Extensive knowledge of **Snowflake Database**, including **database schema** and **table structures**.
- Experienced in evaluating the performance of **LLMs** using metrics such as **accuracy, precision, recall, F1-score, and perplexity**, and adept at interpreting model outputs to assess their quality.
- Preprocessed and cleaned messy data, handled **outliers** and performed **data wrangling** tasks efficiently using **Python libraries** for real-world data analysis.
- **GenAI** assists to evaluate the performance of **Machine Learning (ML Infrastructure)** using various metrics and techniques like **Cross-validation, Hyperparameter Tuning, and Model Comparison** to select the best-performing model for a given task.
- Collaborated with **software developers** that predominantly use **Java** for application development.
- Experience in **fact dimensional modeling (STAR, Snowflake)**, **transactional modeling**, and **SCD**.
- Experienced in documenting **LLM development** processes, experiments, and results, and proficient in communicating technical concepts and findings to both technical and non-technical stakeholders.
- Worked on **Artificial Intelligence (AI) algorithms** to explore and visualize the characteristics of the data.
- Developed **Predictive Analytics** using **PySpark** and **Spark SQL** on **Databricks** to extract, transform and uncover insights from the raw data.
- Scaled **LLM training** and inference tasks using parallel and distributed computing frameworks such as **TensorFlow Distributed Training, PyTorch Distributed Data Parallel (DDP)**, or distributed computing platforms like **Apache Spark** and **Spark SQL**.

Environment: SQL (PostgreSQL, MySQL, Snowflake), Azure Cloud, Azure Data Factory (ADF), Databricks, ADLS GEN2, Azure EC2, GCP ML Libraries, EMR, Hadoop, Hive, Spark RDDs, PySpark, Spark SQL, ElasticSearch, Python, Pandas, NumPy, JSP, JSTL, HTML, CSS, JavaScript, AJAX, Java, TensorFlow, PyTorch (DDP), GPT Models, LLMs, Generative AI (GenAI), NLP, Reinforcement Learning, Random Forest, Time Series Models, Generalized Linear Models, Regularization, Tableau, BusinessObjects, Chart.IO.

Client: Quotient, Mountain View, CA

Oct 2019 – May 2021

Role: Data Scientist

Responsibilities:

- Led efforts in **gathering, analyzing, and translating business requirements** into advanced **data science solutions**, aligning technical strategies with organizational goals.

- Designed and implemented robust **machine learning pipelines** using **Python (NumPy, Pandas, SciPy, Scikit-learn, NLTK, Matplotlib, Seaborn)** and **R** for statistical analysis and predictive modeling.
- Partnered with cross-functional **business and data science platform teams** to resolve challenges related to **security, reliability, performance, and data quality**.
- **Built, tuned, and deployed ML models** both on-premises and in the **AWS Cloud**, leveraging **Amazon ECS, AWS Fargate, S3, DynamoDB, Lambda, and EC2** for scalable, secure deployment.
- Performed **Exploratory Data Analysis (EDA)**, including **descriptive statistics, outlier detection, hypothesis testing, and factor analysis**, ensuring high-quality features for modeling.
- Improved predictive performance by 5% through **ensemble techniques** such as **Bagging, Gradient Boosting, XGBoost, and AdaBoost**.
- Developed **churn prediction models** using **Logistic Regression, Random Forest, and Support Vector Machines (SVM)** to identify at-risk customers and reduce attrition.
- Built **classification models** for targeted promotions, increasing purchase likelihood through **Decision Trees and Logistic Regression**.
- Implemented **data pipelines** with **AWS Glue, AWS Data Pipeline, and PySpark**, automating ML training and inference workflows.
- Conducted **data extraction and wrangling** from **SQL databases, MongoDB, and Excel/Access**, ensuring data readiness for analytics.
- Designed scalable solutions for **unstructured data aggregation and preprocessing** using **BeautifulSoup, Pandas, and NumPy**.
- Monitored **ML models in production** using **AWS CloudWatch and CloudTrail**, ensuring system reliability and real-time performance tracking.
- Created **interactive dashboards and executive reporting tools** with **Power BI, Tableau, Seaborn, and Matplotlib** to drive data-driven decision-making.
- Designed and optimized **SQL scripts and PySpark transformations**, enhancing scalability and reducing latency in analytics workflows.
- Drove the adoption of **cloud-native ML infrastructure** across **AWS, on-prem, and hybrid platforms**, enabling enterprise-grade AI deployment.
- Mentored junior data scientists in **model development, statistical methods, cloud deployment, and business storytelling**, fostering team growth and knowledge sharing.

Environment: SQL, MongoDB, Excel, Access, AWS Cloud (ECS, Fargate, S3, DynamoDB, Lambda, EC2, Glue, Data Pipeline, CloudWatch, CloudTrail), On-prem/Hybrid Platforms, Apache Spark, PySpark, Python, R, NumPy, Pandas, SciPy, Scikit-learn, NLTK, BeautifulSoup, XGBoost, AdaBoost, Random Forest, Support Vector Machines (SVM), Logistic Regression, Decision Trees, Ensemble Techniques (Bagging, Gradient Boosting), Power BI, Tableau, Matplotlib, Seaborn.

Client: TD Bank, Chicago, IL

Sep 2017-Sep 2019

Role: Sr. Data Analyst

Responsibilities:

- Acted as a subject matter expert on **regulatory compliance** matters, providing invaluable guidance and recommendations to stakeholders, ensuring adherence to applicable regulations and standards.
- Developed **Python, Shell, and Hive Scripts** to simplify **data ingestion** from various sources, ensuring smooth data pipelines.
- Developed & Modified **Python Scripts, Shell Scripts, and Hive Scripts** for **Data Ingestion**. Ingested data from various sources.

- Managed **Oozie workflows** with **Sqoop** for automated **data ingestion** into **HDFS**, improving efficiency and scalability.
- Collaborated with **fraud team** to establish **data ingestion workflows**, integrating **mainframe** and **database data** with **ML algorithms** for enhanced **fraud detection**.
- Led the setup of **Hive tables**, aligning with requirements and managing **Static and Dynamic Partitions** for efficient **data management**.
- Designed and implemented custom **ETL processes** using **Python** and **PySpark** to extract, transform, and load data from various sources into **Hadoop tables**, ensuring **data integrity** and **consistency** for **fraud detection analytics**.
- Employed **Agile methodologies (Scrum)** and **project management tools** such as **JIRA** to effectively plan, coordinate, and monitor project activities, ensuring strict adherence to timelines and budget constraints.
- Implemented **project management** practices to mitigate risks, resolve conflicts, and drive **continuous improvement**, resulting in the successful delivery of project objectives.
- Led regular **project status meetings** to provide comprehensive updates on progress, address issues, and solicit feedback, ensuring stakeholders remained informed and engaged.
- Developed detailed **project documentation**, including **project plans, requirements documents, and status reports**, maintaining transparency and accountability across the project team.
- Utilized **Tableau, Power BI, and Excel (Lookup & Pivot Tables)** generating actionable insights to support **data-driven decision-making**.
- Utilized **R, Python, and SQL** for sophisticated **data manipulation, analysis, and modeling**, enabling the development of **predictive analytics models** and advanced reporting solutions.
- Developed **Python-based fraud detection algorithms** and used **PySpark** for **large-scale data processing**, resulting in the identification of **fraudulent activities** within the organization.
- Explored and implemented **supervised learning algorithms** such as **Random Forest, Logistic Regression, and Extreme Gradient Boosting (XGBoost)** algorithms for classification tasks in **fraud detection**, achieving a precision-recall balance that effectively minimized false positives while capturing fraudulent activities accurately.
- Used **k-means clustering** to detect anomalous patterns in **transactional data**, enabling the identification of previously unknown fraud schemes.
- Proficiently crafted **SQL queries** in **Teradata** for **data quality assessment** and developed advanced **stored procedures** for thorough **data profiling**.
- Engineered **Oozie workflows** with **Sqoop, MapReduce, and Shell actions**, enabling seamless **code deployment** through **GIT** and **UDeploy**.
- Developed and implemented a robust **data integration framework** using **Python** and **SQL**, streamlining **ETL processes** and enabling seamless data extraction, transformation, and loading from multiple sources.
- Utilized **Python** on **Edge nodes** for **data querying operations** via **Hive and Impala**, enhancing **data processing capabilities**.

Environment: SQL (Teradata), Hadoop, HDFS, Hive, Impala, Sqoop, Oozie, MapReduce, PySpark, Python, Shell Scripting, R, Machine Learning (XGBoost, Random Forest, Logistic Regression, K-means Clustering), ETL Pipelines, Data Profiling, Tableau, Power BI, Excel, Git, UDeploy, JIRA, Agile (Scrum).

Client: WNS Global Services, Bengaluru, India

Aug 2013 - June 2016

Role: Python Developer

Responsibilities

- Involved in **requirement gathering and business analysis**, translating business needs into **Python-based data solutions** for Hadoop and Big Data environments.
- Designed and implemented **database schemas** and developed **Python scripts** to create tables, perform **data manipulation**, and manage large-scale datasets.

- Developed, maintained, and optimized **SQL queries and Python-based ETL pipelines**, ensuring data accuracy and reliability.
- Built **Python modules for data cleaning, preprocessing, and transformation**, converting massive raw datasets into structured and insightful formats.
- Created **interactive dashboards and automated reports** using **Python (Plotly, Matplotlib, Seaborn)** integrated with **Tableau and Power BI** for organizational decision-making.
- Automated **invoice record management** and data validation tasks using **Python scripting**, improving efficiency and accuracy in financial workflows.
- Performed **data extraction, feature engineering, and validation** using **Pandas, NumPy, and SciPy**, drawing meaningful insights for retail and marketing products.
- Conducted **Proof of Concepts (POCs)** with Python frameworks to evaluate new data analytics and visualization approaches.
- Collaborated with **cross-functional teams** and conducted **peer code reviews** to maintain quality, performance, and correctness of Python applications.
- Automated manual business processes with **Python scripts**, developing reusable tools to enhance team scalability and productivity.
- Designed and executed **Python queries and APIs** to retrieve, transform, and integrate data from multiple sources for real-time business insights.
- Developed **custom data visualization solutions** in Python, building charts, graphs, and tables to communicate KPIs effectively.
- Built **Python-based analytics tools** for performance optimization, improving team scalability and profitability.

Environment: SQL, Hadoop, Big Data, Database Schema Design, Python (Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly), ETL Pipelines, Data Manipulation, Data Validation, API Integration, Proof of Concepts (POCs), Tableau, Power BI, Business Analysis, Automation Scripting.